



MSc Group Design Specification 2025

CADEM0016

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Revision History

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Contents

1	Introduction	2
2	The Design Task	2
2.1	The Cargo	2
2.2	Reference Flights - (2026 season)	3
2.2.1	Delivery Timings	3
2.3	Key Design Drivers	5
3	Design Specification	5
3.1	Aircraft Variant	5
3.2	Airport Compatibility	5
3.3	Design Payload	6
3.4	Design Missions	6
3.5	Performance Requirements	6
3.6	Propulsion and Fuel	7
3.7	Cabin Interior	7
3.8	Guidelines / Other Constraints	7
4	Economic Evaluation	7
4.1	Crew Salaries	8
4.2	Landing Fees	8
4.3	Parking Fees	8
4.4	Fuel Costs	8
4.5	Maintenance and Insurance Costs	8
A	Nominal Mission Profile	9
A.1	Taxi	10
A.2	Take-off, Initial Climb to 1500ft, Acceleration to 250 kts	10
A.3	Climb	10
A.4	Cruise	10
A.5	Descent	11
A.6	Landing	11
A.7	To Gate	11
A.8	Alternate Mission	11
A.8.1	Loiter	11
A.8.2	Contingency Fuel	12
B	Baseline Aircraft/Fleet	12

1 Introduction

Formula 1 (F1) transports over 1200 tonnes of equipment to each race on the calendar. For each of the fly-away races, critical components such as car components, timing systems and cameras must be transported by air freight.

Formula 1 aims for its operations to be carbon net-zero by 2030, and air transport accounts for a significant proportion of these emissions. To help mitigate these emissions, Formula 1 has commissioned a feasibility study to develop its own fleet of cargo aircraft to reduce emissions. This report outlines the parameters of this feasibility study.

2 The Design Task

This task is to complete the *conceptual* design of a custom cargo aircraft for entry into service in 2040. A fleet of these aircraft will be used to transport the cargo specified in 2.1 across each 'flyaway' leg of a Formula One (F1) season. The 2026 F1 calendar will serve as the reference season, and the required flights are detailed in 2.2.

The design should aim to minimise both the direct operating cost and climate impact of the required flights over the course of a season.

2.1 The Cargo

It is assumed that 12 teams will participate in Formula 1 in 2040. For each leg in which air cargo is required, it is assumed that each team will transport

- 13 P6P pallets, each with a mass of 3 tonnes

In addition to the teams, Formula 1 itself must also transport equipment by air freight, such as the safety cars, timing systems and communication systems, etc. It is assumed F1 will transport:

- four P6P+ pallets, each with a mass of 2.5 tonnes (the safety / medical cars...)
- 86 P6P pallets, each with a mass of 3 tonnes

The dimensions of each type of pallet are shown in Fig. 1 and Table 2.

Table 2: Unit Load device dimensions

Unit Load Device	Width [cm]	Length [cm]	Height [cm]
P6P	243.5	317.5	162.6
P6P+	243.5	497.8	162.6

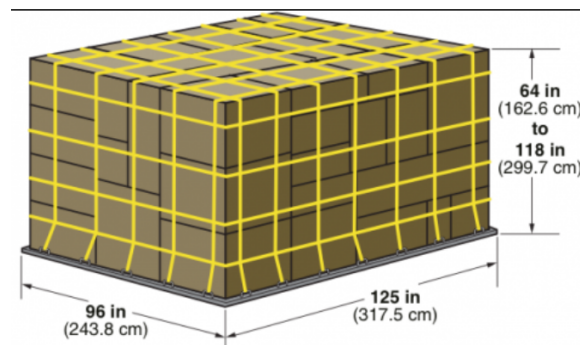


Figure 1: Dimensions of standard P6P shipping unit.

2.2 Reference Flights - (2026 season)

The 2026 Formula One calendar (Fig. 2) will be used as a reference season for the design task. This season consisted of 24 races spaced between March and December. However, not all races require air freight. Table 3 details if airfreight is needed in/out of each race. In some cases, freight can be driven to the next race, and the fleet of aircraft will fly empty to the next destination (see notes for races 6,8 and 19).

2.2.1 Delivery Timings

Cargo must arrive at the venue by 6 am on the Wednesday of a race week. If race weekends are 'back-to-back' (e.g. 7 days between races), then the delivery schedules for freight can become quite tight (see Fig. 3). It is assumed that:

- the last cargo aircraft takes off 40 hours after the start of the race
- cargo must arrive at the venue by 6 am on the Wednesday of a race week
- Between an aircraft landing and cargo arriving at a circuit takes six hours
- The aircraft is stationary for two hours, if a mid-flight refueling stop is required

Ensure you account for the timezone changes in your analysis!

 MAR 06-08 ROUND 1 MELBOURNE AUSTRALIA	 MAY 22-24 ROUND 7 MONTREAL CANADA	 JUL 24-26 ROUND 13 BUDAPEST HUNGARY	 OCT 23-25 ROUND 19 AUSTIN UNITED STATES
 MAR 13-15 ROUND 2 SHANGHAI CHINA	 JUN 05-07 ROUND 8 MONACO MONACO	 AUG 21-23 ROUND 14 ZANDVOORT NETHERLANDS	 OCT NOV 30-01 ROUND 20 MEXICO CITY MEXICO
 MAR 27-29 ROUND 3 SUZUKA JAPAN	 JUN 12-14 ROUND 9 BARCELONA - CATALUNYA SPAIN	 SEP 04-06 ROUND 15 MONZA ITALY	 NOV 06-08 ROUND 21 SÃO PAULO BRAZIL
 APR 10-12 ROUND 4 SAKHIR BAHRAIN	 JUN 26-28 ROUND 10 SPIELBERG AUSTRIA	 SEP 11-13 ROUND 16 MADRID SPAIN	 NOV 19-21 ROUND 22 LAS VEGAS LAS VEGAS
 APR 17-19 ROUND 5 JEDDAH SAUDI ARABIA	 JUL 03-05 ROUND 11 SILVERSTONE GREAT BRITAIN	 SEP 24-26 ROUND 17 BAKU AZERBAIJAN	 NOV 27-29 ROUND 23 LUSAIL QATAR
 MAY 01-03 ROUND 6 MIAMI MIAMI	 JUL 17-19 ROUND 12 SPA-FRANCORCHAMPS BELGIUM	 OCT 09-11 ROUND 18 SINGAPORE SINGAPORE	 DEC 04-06 ROUND 24 YAS ISLAND ABU DHABI

Figure 2: The Formula 1 2026 Calendar.

Table 3: The Formula 1 2026 Air freight requirements.

#	Race Name	City	Date (2026)	Race Time (Local)	Closest Airport	Air Freight In?	Air Freight Out?	Notes
1	Australia	Melbourne	March 6-8	15:00	MEL	yes	yes	pre-race fleet flies in from LHR
2	China	Shanghai	March 13-15	15:00	PVG	yes	yes	
3	Japan	Suzuka	March 27-29	14:00	NGO	yes	yes	
4	Bahrain	Sakhir	April 10-12	18:00	BAH	yes	yes	
5	Saudi Arabia	Jeddah	April 17-19	20:00	JED	yes	yes	
6	USA	Miami	May 1-3	16:00	MIA	yes	no	Fleet flies empty to Montreal
7	Canada	Montreal	May 22-24	14:00	YUL	no	yes	
8	Monaco	Monaco	June 5-7	15:00	NCE	yes	no	Fleet flies empty to Madrid
9	Spain	Barcelona-Catalunya	June 12-14	15:00	BCN	no	no	
10	Austria	Spielberg	June 26-28	15:00	VIE	no	no	
11	Great Britain	Silverstone	July 3-5	15:00	LHR	no	no	
12	Belgium	Spa-Francorchamps	July 17-19	15:00	BRU	no	no	
13	Hungary	Budapest	July 24-26	15:00	BUD	no	no	
14	Netherlands	Zandvoort	Aug 21-23	15:00	AMS	no	no	
15	Italy	Monza	Sept 4-6	15:00	MPX	no	no	
16	Spain	Madrid*	Sept 11-13	15:00	MAD	no	yes	
17	Azerbaijan	Baku	Sept 24-26	15:00	GYS	yes	yes	
18	Singapore	Singapore	Oct 9-11	20:00	SIN	yes	yes	
19	USA	Austin	Oct23-25	14:00	AUS	yes	no	Fleet flies empty to Mexico
20	Mexico	Mexico City	Oct 30 - Nov 1	14:00	MEX	no	yes	
21	Brazil	Sao Paulo	Nov 6-8	14:00	GRU	yes	yes	
22	USA	Las Vegas	Nov 19-21	22:00	LAS	yes	yes	The race occurs on Saturday night
23	Qatar	Lusail	Nov 27-29	20:00	DOH	yes	yes	
24	Abu Dhabi	Yas Marina	Dec 4-6	17:00	AUH	yes	yes	post-race fleet flies to the LHR

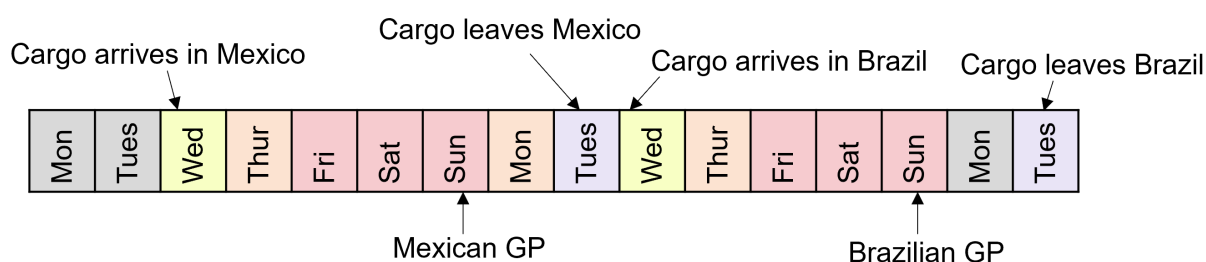


Figure 3: Example Cargo timeline between the Mexican and Brazilian races.

2.3 Key Design Drivers

The final aircraft design shall be chosen to meet the specific requirements of the given mission, whilst minimising both the direct operating cost (DOC) and environmental impact.

Section 3 details constraints on the aircraft design. Section 4 and ?? provide guidelines for how the DOC and climate impact of an aircraft can be evaluated, respectively.

Specific considerations designers may choose to consider include (in no particular order):

- Aircraft maximum payload mass. The total cargo may be carried over multiple aircraft; feasible solutions may range from a single aircraft to a fleet of 10's to 100's of aircraft. Trade-offs across the possible range of fleet sizes must be considered.
- The use of ground folding wingtips (see Section 3.2) enables significant increases in wingspan when compared to current aircraft. The use of suitable tools to estimate the mass and performance of such designs should be considered.
- The chosen cruise Mach number and altitude will have a dramatic effect on aircraft planform, performance, and climate impact.

As detailed in Section B, chosen designs should be compared to the B777F.

3 Design Specification

3.1 Aircraft Variant

A single aircraft variant capable of transporting the specified payload is required. No future potential for family development needs to be considered in the design philosophy or major component sizing. No consideration of other potential uses or other payloads is required.

The size of the aircraft fleet is to be chosen by the designers, and will remain constant across the season.

3.2 Airport Compatibility

In the Landing and Take-off configuration (e.g., configuration from runway-to-runway), you may assume all airports listed in Table 3 can accept aircraft up to code 'F' (see Table 4).

In the taxi configuration (e.g. runway-to-gate), you may assume all airports listed in Table 3, can accept aircraft up to code 'E' (see Table 4).

Table 4: ICAO Aerodrome Reference Code (Letter) – Maximum Wingspan and Main Landing Gear Span

Code Letter	Maximum Wingspan Range	Main Landing Gear Span Range
A	up to 15 m	up to 4.5 m
B	15 m but < 24 m	4.5 m but < 6 m
C	24 m but < 36 m	6 m but < 9 m
D	36 m but < 52 m	9 m but < 14 m
E	52 m but < 65 m	9 m but < 14 m
F	65 m but < 80 m	14 m but < 16 m

3.3 Design Payload

The fleet of aircraft should be capable of carrying the cargo as detailed in Section 2.1 as well as four crew members.

At the conceptual design stage, the following assumptions are deemed acceptable:

- The centre of mass of the total payload may be assumed constant across all payload mass levels. E.g. at 25%, 50% or 100% of the maximum payload mass, variations in the centre of mass of the combined payload do not need to be considered.

3.4 Design Missions

The fleet of aircraft must be able to complete the missions detailed in Section 2.2, using the mission profile specified in Section A.

If a mid-flight refuelling stop is required, each mission may be split into a series of shorter 'legs', in which case each 'leg' of a mission must also use the mission profile specified in Section A.

3.5 Performance Requirements

The aircraft shall be design in accordance with FAR-25/CS-25 regulations.

All technical requirements shall be set to meet ETOPS reliability standards. Airlines shall be capable of applying for certification to fly 180 minutes ETOPS operations at service entry.

The aircraft shall be designed in accordance with design speeds set out in Section 3.5.

Table 5: Structural design speed and Mach definitions

Parameter	Definition
Structural design speed V_c	350 kt CAS
Design dive speed V_D	$V_c + 35$ kt CAS
Design cruise Mach M_c	$M_{\text{cruise}} + 0.04$
Design dive Mach M_D	$M_c + 0.03$ (assumes Fly-by-wire (FBW) with full envelope protection)

3.6 Propulsion and Fuel

The aircraft should be powered by conventional off-the-shelf kerosene/drop-in-SAF fuelled engines, either turbofan or turboprop. The number and type of engines is not specified. The density of kerosene or SAF is assumed to be 0.8kg per litre.

3.7 Cabin Interior

Flight Deck to have 2 flight crew operation. The additional cabin area should include four simple seats to allow carrying spare crew, a simple lavatory and space for crew baggage and informal catering. Maximum cockpit and cabin area pressure altitude = 8000ft at maximum cruise altitude.

3.8 Guidelines / Other Constraints

- Novel concepts and configurations are not excluded. Any chosen concept/configuration (conventional or novel) must be justified.
- Allowances for technologies appropriate for the expected entry into service (EIS) date of 2040 may be included, but their suitability, application and benefits must be justified.
- The initial specification should be seen as the starting point. Requirements may change in the early stages of any project due to feedback from the market or aircraft design studies. If you believe parts of the specification are too restrictive, or should be altered, please consult the Unit Director.

4 Economic Evaluation

The following guidelines can be used to estimate the Direct Operating Costs (DOC) per year of operation. These simplified values are intended to fairly penalise larger and heavier aircraft, so that the choice of fleet size becomes a challenging design trade-off.

4.1 Crew Salaries

Each aircraft is assumed to operate with two flight crew and two relief crew for long-haul ETOPS operations. The annual salary per crew member is estimated at \$150,000, giving a total of approximately \$600,000 per aircraft per year.

4.2 Landing Fees

Landing fees are assumed to be charged as a linear function of Maximum Take-Off Mass (MTOM). For simplicity, a constant rate of \$25 per tonne of MTOM is applied at all airports.

- Example: 100 t aircraft → \$2,500 per landing
- Example: 400 t aircraft → \$10,000 per landing

4.3 Parking Fees

Parking fees are assumed to be charged per day, based on the ICAO Aerodrome Reference Code (Letter). All airports are assumed to apply the same daily rates:

ICAO Code Letter	Wingspan Range	Parking Fee (per day)
C	24–36 m	\$1,000
D	36–52 m	\$2,000
E	52–65 m	\$4,000
F	65–80 m	\$6,000

These rates are charged based on the aircraft's ground span in the 'taxi' configuration. During a "re-fuelling" stop, these prices may be divided by 24 to get the hourly rate.

4.4 Fuel Costs

For the purposes of this study, the following simplified assumptions are made regarding fuel prices in 2040. These fixed values provide a consistent basis for Direct Operating Cost (DOC) estimation.

- Jet-A1 kerosene is assumed to cost **\$1.00 per litre**.
- Sustainable Aviation Fuel (SAF) is assumed to cost **\$2.00 per litre**.

These values are applied uniformly across all missions and airports in the reference season.

4.5 Maintenance and Insurance Costs

For this conceptual study, both maintenance and insurance costs are assumed to scale with the aircraft's hull value. The hull value itself is represented using a simple empirical model

based on maximum take-off mass (MTOM), reflecting the general trend that larger aircraft have higher acquisition and upkeep costs.

The hull value is estimated as:

$$V_{\text{hull}} = 44,880 \text{ MTOM}^{0.65} \quad (1)$$

where MTOM is the maximum take-off mass in kilograms and V_{hull} is the estimated hull value in dollars.

Maintenance Costs

Maintenance costs are represented using a fixed annual component and a variable component that scales with aircraft utilisation. Both components are assumed to be proportional to the aircraft's hull value:

$$C_{\text{m, fixed}} = 0.03 V_{\text{hull}} \quad (2)$$

$$C_{\text{m, var}} = 5 \times 10^{-6} V_{\text{hull}} \quad [\$/\text{FH}] \quad (3)$$

where $C_{\text{m, fixed}}$ is the fixed maintenance cost per year (in dollars) and $C_{\text{m, var}}$ is the cost per flight hour, in dollars.

Insurance Costs

Annual insurance premiums are assumed to be a fixed fraction of the hull value, representing a simplified combination of hull and liability coverage:

$$C_{\text{ins}} = 0.005 V_{\text{hull}} = 224.4 \text{ MTOM}^{0.65} \quad (4)$$

Here C_{ins} is the estimated annual insurance premium in dollars, corresponding to a nominal rate of 0.5% of the hull value.

A Nominal Mission Profile

Any flight undertaken by the aircraft must follow the mission profile outlined in Fig. 4, which adheres to the EASA Rule for Air Operations [1]. This mission consists of various flight phases, which are detailed in the following sub-section.

The mission profile is split into two distinct sections: the 'mission', and the 'alternate', where the alternate route simulates a diversion to a secondary airport. The length of the mission range is chosen to match the flight specifications, whereas the alternate range is fixed at 350km¹

¹Unlike other parameters, the alternate range is not specified in the regulations, as the distance to the alternate airport is tailored for each flight plan. However, for this project, a constant value is assumed for simplicity.

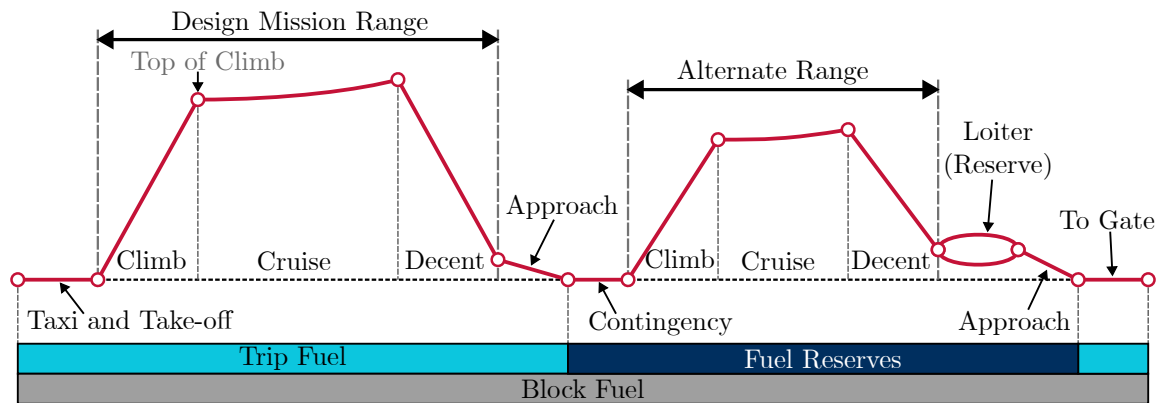


Figure 4: Design mission segment and fuel breakdown

A.1 Taxi

Aircraft taxis to the runway.

Time: 20 minutes

Fuel: 20 minutes at idle power

A.2 Take-off, Initial Climb to 1500ft, Acceleration to 250 kts

The aircraft must be capable of a balanced field take-off (brakes-off to 35ft above the ground) at all airports specified in Table 3 at maximum take-off mass in ISA+15C conditions.

Time: 1 minute

Fuel: can assume 1 minute at maximum power

A.3 Climb

Following take-off as above, the aircraft must be capable of a time to climb (TTC) from 1500ft to an Initial Cruise Altitude (ICA) of at least 20000ft in 30 minutes or less in ISA conditions. Speed below 10000 ft should not exceed 250kts CAS. Speed/Mach profile above 10000ft should not exceed the chosen cruise speed.

A.4 Cruise

The aircraft will cruise at constant calibrated airspeed or Mach number. The altitude of the aircraft can follow one of three profiles during cruise:

- A constant flight level (e.g. 20000ft, 22000ft, 24000ft...).
- A stepped cruise-climb if beneficial for the mission.

- Early in the design process, a constant C_L cruise can be assumed with appropriate contingency factors.

The following conditions must be met at the chosen cruise altitude

- Must be greater than or equal to 20,000ft
- Less than or equal to the operational ceiling of the aircraft
- Buffet margin ≥ 0.3 (cruise Mach number or speed)
- Level flight possible with maximum cruise thrust (cruise Mach number or speed)
- Vertical climb rate ≥ 300 ft/min with maximum climb setting and at climb Mach number or speed

A.5 Descent

Descent from cruise should be carried out in ISA conditions. It is normal for the descent speed profile with altitude to be similar to the climb speed profile.

A.6 Landing

The aircraft must be able to land at all airports specified in Table 3 at maximum landing weight in ISA conditions with an approach speed of less than or equal to 145 kts.

A.7 To Gate

Aircraft taxis to the gate.

Time: 20 minutes

Fuel: 20 minutes at idle power

A.8 Alternate Mission

For simplicity, the alternate mission range is 350km, and follows the same pattern as the mission profile. With the additions of Contingency and loiter fuel.

A.8.1 Loiter

Before the final landing, the aircraft must loiter for a minimum of 30 minutes at the minimum drag velocity at 1500ft in ISA conditions.

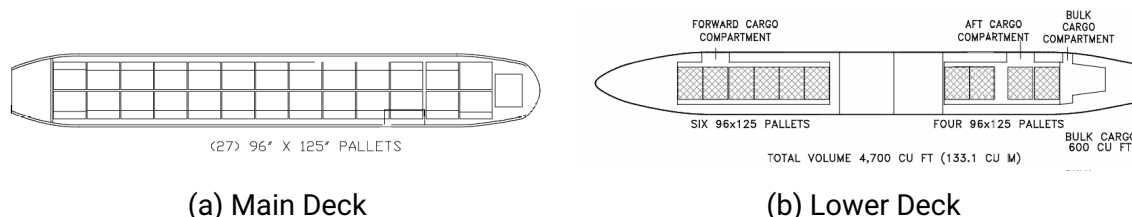


Figure 5: Example Deck Layouts for the B777F with 96x125 inch pallets

A.8.2 Contingency Fuel

As defined in CS-25, contingency fuel is the greater of 5 minutes of loiter time or 3% of the trip fuel.

B Baseline Aircraft/Fleet

The Boeing 777F should serve as a baseline aircraft for comparing the chosen designs. Critical Top-level parameters of the B777F are shown in the Table 6, with an example payload range diagram shown in Section B. The B777F is capable of carrying upto 103 tonnes of payload spread across 27 LD-7 container on the main deck with a further 13 on the lower deck (see Fig. 5).²

Table 6: Critical Mass Values for Boeing 777F

Parameter	Mass [tonnes]
Maximum Take-Off Mass (MTOM)	348.7
Maximum Landing Mass (MLM)	260.8
Maximum Payload	103.0
Operational Empty Mass (OEM)	145.0
Maximum Fuel Mass	145.6

The freight outlined in Section 2.1 totals 246 pallets and 736 tonnes of payload. In practice, this means the problem is mass-constrained (rather than volume-constrained), requiring 8 B777Fs, each carrying 31 pallets and 92 tonnes of freight.

The developed conceptual design framework should be capable of replicating a B777F-like aircraft to allow a fair comparison of DOC and climate impact.

References

- [1] Anon. Easy access rules for air operations — revision 21. Technical report, Agency, European Union Aviation Safety, 2023.

²All information for the B777F sourced from aircraft airport characteristics document <https://www.boeing.com/commercial/airports/plan-manuals> [accessed 12/12/2025]

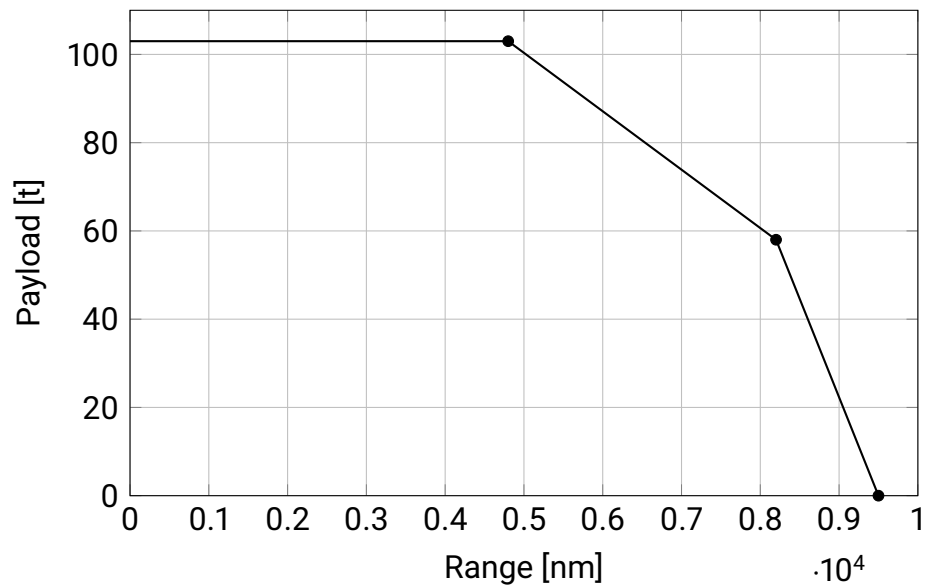


Figure 6: Payload–Range diagram for Boeing 777F.